BAIT AL MANDI — System Documentation

BAIT AL MANDI

System Architecture Documentation

بيت المندي — وثيقة البنية المعمارية للنظام

Version 1.0

June 15, 2026

Name:	Bait Al Mandi Restaurant
Project:	Full-Stack Restaurant Management System
Tech Stack:	Next.js 14 · Supabase · TypeScript
Version:	1.0
Created:	June 15, 2026
Exported:	June 15, 2026
Classification:	Confidential

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Section 1: System Overview

1.1 What is Bait Al Mandi?

A full-stack restaurant management system for **Bait Al Mandi** — a Yemeni restaurant in **Sanaa, Yemen**. The system provides:

- **Public Website** (Landing Page, Menu, Cart, Ordering, Tracking)
- **Admin Dashboard** (Dashboard, Orders, Menu CRUD, Categories, Offers, Gallery, Reviews, Reports, Delivery Settings, Site Settings)
- **Delivery System** (Distance-based pricing, weather/peak surcharges, OSRM Routing, interactive map)
- **Invoice System** (PNG, PDF, direct print, QR Code for tracking)
- **Offer & Bundle System** (Pricing Engine: fixed price, percentage discount, amount discount, free item)
- **Order Tracking System** (Unique Tracking Token per customer, QR Code)
- **Reports & Analytics** (12 API routes + interface tabs + print + Excel)

1.2 Problem & Solution

Problem	Solution
No digital restaurant management system	Complete system from ordering to delivery to reports
Delivery without maps or fair pricing	OSRM Routing + distance-based pricing + weather/peak surcharges
Traditional paper invoices	PNG/PDF/print invoices with QR tracking
Complex offers without a pricing engine	Pricing Engine with 4 offer types
Order tracking difficulty	Tracking Token + tracking page per customer

1.3 Tech Stack

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- **Framework:** Next.js 14.2.3 (App Router)
- **Language:** TypeScript 5.4.5
- **Styling:** Tailwind CSS + CSS Modules
- **Database:** Supabase (PostgreSQL 15)
- **ORM:** Drizzle ORM + Supabase REST (dual access)
- **Auth:** Supabase Auth (email/password) + Row Level Security
- **State (Client):** Zustand (cartStore) + React Context (Settings)
- **Charts:** Recharts
- **Maps:** Leaflet + OSRM (routing)
- **GIS:** Turf.js + RBush (R-Tree spatial indexing)
- **PDF/PNG:** jsPDF + html2canvas + QRCode.react
- **Excel:** ExcelJS
- **Animations:** Framer Motion + GSAP
- **Email:** Nodemailer

1.4 Key Versions

Package	Version
Next.js	`^14.2.3`
React	`^18.3.1`
Supabase SSR	`^0.5.1`
Drizzle ORM	`^0.30.10`
Zustand	`^5.0.0`

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Section 2: Directory Structure

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```
baitalmandiwibapp/
├── src/
│   ├── actions/          # Server Actions (RSC mutations)
│   │   ├── categories.ts     # Categories CRUD
│   │   ├── items.ts          # Items & Prices CRUD
│   │   ├── orders.ts         # Order creation + delivery calc + status update
│   │   └── orders-offers.ts  # Fetch offer info for order
│   │
│   ├── app/              # App Router (pages + API)
│   │   ├── page.tsx          # Landing Page
│   │   ├── layout.tsx        # Root Layout (Navbar, Footer, SettingsProvider)
│   │   ├── globals.css       # Global styles (Custom Properties)
│   │   ├── page.module.css   # CSS Module for landing page
│   │   ├── favicon.ico
│   │   ├── menu/page.tsx     # Menu page (view + search + cart)
│   │   ├── cart/page.tsx     # Cart page (legacy)
│   │   ├── my-orders/page.tsx # Cart + map + full order flow
│   │   ├── contact/page.tsx  # Contact page + review submission
│   │   ├── gallery/page.tsx  # Photo gallery
│   │   ├── test-map/page.tsx # Map test page
│   │   ├── t/[token]/page.tsx# Order tracking by token
│   │   ├── track-order/[orderId]/page.tsx # Order tracking by UUID
│   │   ├── admin/            # Admin panel
│   │   │   ├── layout.tsx    # Admin Layout (Sidebar + Auth)
│   │   │   ├── page.tsx      # Admin Dashboard
│   │   │   ├── login/        # Login page + actions
│   │   │   ├── orders/       # Order management
│   │   │   ├── menu/         # Menu CRUD
│   │   │   ├── categories/   # Categories CRUD
│   │   │   ├── offers/       # Offers CRUD
│   │   │   ├── gallery/      # Gallery management
│   │   │   ├── reviews/      # Reviews management
│   │   │   ├── reports/      # Reports (9 tabs)
│   │   │   ├── delivery/     # Delivery settings
│   │   │   └── settings/     # Site settings
│   │   └── api/             # API Routes
│   │       ├── auth/login/route.ts
│   │       ├── resolve-zone/route.ts (legacy)
│   │       └── reports/ (12 endpoints)
│   │
│   ├── components/
│   │   ├── admin/reports/    # 9 report tab components
│   │   ├── invoice/
│   │   │   ├── InvoiceModal.tsx    # Invoice modal (PNG/PDF)
│   │   │   └── receipt-html.ts    # Receipt HTML for printing
│   │   └── layout/          # Navbar, Footer, LoadingScreen
│   │
│   ├── db/
│   │   ├── index.ts         # Drizzle ORM initialization
│   │   ├── schema.ts        # Complete schema (22 tables)
```


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```
│   │   └── rls.sql          # Row Level Security policies
│   │
│   ├── lib/
│   │   ├── supabase.ts      # Supabase Client (Browser)
│   │   ├── permissions.ts   # Permission system (3 roles)
│   │   ├── constants.ts     # Constants (contact info)
│   │   ├── settings-context.tsx # React Context for settings
│   │   ├── delivery-pricing.ts  # Single source for delivery fee calc
│   │   ├── delivery-routing.ts  # OSRM Routing + fallback
│   │   ├── delivery-zones.ts    # (Legacy) zone system
│   │   ├── offer-pricing.ts     # Offer pricing engine
│   │   ├── bundle-utils.ts      # Bundle info extraction
│   │   ├── location-validation.ts # Sanaa location validation
│   │   ├── exportExcel.ts       # Excel export
│   │   ├── printReport.ts       # Report printing
│   │   └── geo/                 # (Legacy) zone resolution
│   │
│   ├── middleware.ts         # Next.js Middleware (Admin Routes)
│   ├── types.d.ts            # Global TypeScript types
│   ├── services/reports/     # Report engine (Drizzle-based)
│   └── utils/supabase/       # Supabase utilities
│
├── supabase/migrations/      # 11 SQL Migrations (0000-0010)
├── sanaa-map-integration/    # GeoJSON data for Sanaa
├── drizzle.config.ts
├── next.config.js
├── tailwind.config.ts
├── tsconfig.json
├── package.json
└── .env.local
```


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Section 3: Database Architecture

3.1 Core Tables (22 Tables)

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#	Table	Description	Key Fields
1	`users`	Restaurant customers	`id` (PK), `full_name`, `phone` (UK)
2	`admin_users`	Admin accounts	`id` (PK), `email` (UK), `role` (enum)
3	`categories`	Menu categories	`id` (PK), `slug` (UK), `sort_order`
4	`items`	Menu items	`id` (PK), `category_id` (FK)
5	`item_prices`	Item price variants	`id` (PK), `item_id` (FK), `original_price`
6	`offers`	Discount offers/bundles	`id` (PK), `offer_type`, `discount_percent`
7	`offer_items`	Bundle components	`id` (PK), `offer_id` (FK), `menu_item_id` (FK)
8	`cart_sessions`	Cart sessions (legacy)	`id` (PK), `session_id` (UK)
9	`cart_items`	Cart line items (legacy)	`id` (PK), `session_id` (FK)
10	`order_sequences`	Order number sequences	`id` (PK), `sequence_date` (UK)
11	`orders`	**Core table**	`id` (PK), `order_number` (UK), 28 columns
12	`order_items`	Order line items	`id` (PK), `order_id` (FK)
13	`order_status_history`	Status change audit trail	`id` (PK), `order_id` (FK)
14	`order_offers`	Offer snapshot at order	`id` (PK), `order_id` (FK, CASCADE)
15	`order_offer_items`	Bundle lines snapshot	`id` (PK), `order_offer_id` (FK, CASCADE)
16	`gallery_images`	Photo gallery	`id` (PK), `image_url`

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17	`reviews`	Customer reviews	`id` (PK), `rating`
18	`site_settings`	**Key-value settings store**	`id` (PK), `setting_key` (UK), `value` (jsonb)
19	`branches`	Restaurant branches	`id` (PK), `name_ar/en`
20	`audit_logs`	Admin audit logs	`id` (PK), `entity_id`
21	`customer_tokens`	Phone to token mapping	`phone` (PK), `tracking_token`
22	`scheduled_reports`	Scheduled email reports	`id` (PK), `period` (enum)

3.2 Enums (7)

```
admin_role       → 'developer' | 'manager' | 'order_manager'
offer_status     → 'active' | 'expired' | 'disabled'
order_method     → 'whatsapp' | 'website'
order_status     → 'pending' | 'confirmed' | 'preparing' | 'on_the_way' |
'delivered' | 'cancelled'
payment_method   → 'cash' | 'transfer' | 'wallet'
audit_action     → 'modify' | 'cancel' | 'status_change' | 'other'
report_period    → 'daily' | 'weekly' | 'monthly'
```

3.3 Orders Table Structure

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```
orders (
  id                          UUID PK DEFAULT gen_random_uuid()
  order_number                TEXT UK          ← "BAM-YYYYMMDD-XXXX"
  customer_id                 UUID FK→users
  customer_name               TEXT NOT NULL
  customer_phone              TEXT NOT NULL
  tracking_token              TEXT              ← UUID for tracking
  delivery_address            TEXT NOT NULL     ← Required (≥10 chars)
  subtotal                    NUMERIC NOT NULL  ← Server-calculated
  delivery_fee                NUMERIC DEFAULT 0
  tax_amount                  NUMERIC DEFAULT 0
  total_amount                NUMERIC NOT NULL  ← Server recalculation
  notes                       TEXT
  estimated_time              TIMESTAMPTZ
  order_method                order_method NOT NULL
  payment_method              payment_method DEFAULT 'cash'
  offer_id                    UUID FK→offers
  status                      order_status DEFAULT 'pending'
  version                     INTEGER DEFAULT 1  ← Optimistic Locking
  created_at / updated_at     TIMESTAMPTZ
  is_archived / archived_at   BOOLEAN + TIMESTAMPTZ
  is_deleted / deleted_at     BOOLEAN + TIMESTAMPTZ

  -- Delivery Geo Fields
  delivery_latitude           NUMERIC
  delivery_longitude          NUMERIC
  delivery_zone               TEXT              ← null (resolved)
  delivery_distance_km        NUMERIC
  delivery_duration_minutes   INTEGER
  location_verified           BOOLEAN DEFAULT false

  -- Delivery Fee Snapshot Fields (migration 0010)
  base_delivery_fee_amount    NUMERIC DEFAULT 0
  extra_distance_km           NUMERIC DEFAULT 0
  extra_fee_amount            NUMERIC DEFAULT 0
  weather_fee_amount          NUMERIC DEFAULT 0
  peak_fee_amount             NUMERIC DEFAULT 0
  peak_percentage_used        NUMERIC DEFAULT 0
)
```

3.4 Site Settings (Key-Value Store)

The `site_settings` table stores all configurable parameters:

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Key	Type	Default	Description
`restaurant_name`	string	"بيت المندي"	Restaurant name
`restaurant_image`	string	""	Restaurant logo URL
`restaurant_lat`	string	"15.360..."	Restaurant latitude
`restaurant_lng`	string	"44.174..."	Restaurant longitude
`base_delivery_fee`	string	"400"	Base delivery fee
`included_distance_km`	string	"2"	Included distance (km)
`extra_fee_per_km`	string	"100"	Extra fee per km
`max_delivery_distance_km`	string	"15"	Max delivery distance
`road_factor`	string	"1.5"	Road factor (Haversine→Road)
`enable_weather_fee`	string	"false"	Enable weather surcharge
`weather_fee`	string	"200"	Weather fee amount
`enable_peak_hours`	string	"false"	Enable peak hour surcharge
`peak_start_time`	string	"12:00"	Peak start time
`peak_end_time`	string	"14:00"	Peak end time
`peak_percentage`	string	"20"	Peak surcharge percentage
`peak_days`	string	"[]"	Peak days (JSON array)

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Section 4: Pages

4.1 Public Pages

Route	Component	Type	Function
`/`	`page.tsx`	Client	Landing Page: Hero, Menu, Stats, Reviews, Gallery, WhatsApp
`/menu`	`page.tsx`	Client	Full menu with search, categories, cart, offers
`/my-orders`	`page.tsx`	Client	Cart + map + delivery calc + order creation
`/cart`	`page.tsx`	Client	Legacy cart page
`/gallery`	`page.tsx`	Client	Photo gallery with Lightbox
`/contact`	`page.tsx`	Client	Contact form + review + restaurant info
`/t/[token]`	`page.tsx`	Client	Order tracking by token with InvoiceModal
`/track-order/[orderId]`	`page.tsx`	Client	Legacy tracking by UUID
`/test-map`	`page.tsx`	Client	Map and OSRM test page

4.2 Admin Pages

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Route	Component	Function
`/admin`	`page.tsx`	Dashboard: totals, revenue, active items
`/admin/login`	`page.tsx`	Admin login (Supabase Auth)
`/admin/orders`	`page.tsx`	Order management (status updates, details)
`/admin/menu`	`page.tsx`	Menu items CRUD with images and prices
`/admin/categories`	`page.tsx`	Categories CRUD
`/admin/offers`	`page.tsx`	Offers CRUD (4 pricing types)
`/admin/gallery`	`page.tsx`	Gallery image management
`/admin/reviews`	`page.tsx`	Review management (approve/reject)
`/admin/reports`	`page.tsx`	Reports (9 tabs with print/export)
`/admin/delivery`	`page.tsx`	Delivery settings (distance, weather, peak)
`/admin/settings`	`page.tsx`	Site settings (restaurant, contact, bank)

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Section 5: API Routes

5.1 Public Routes

Route	Method	Function
`/api/auth/login`	POST	Admin login (Supabase Auth + Cookies)
`/api/resolve-zone`	POST	Legacy zone resolution from coordinates

5.2 Report Routes (All GET)

Route	Function
`/api/reports/dashboard`	Today's sales, order count, last 10 orders
`/api/reports/orders`	Order logs with filtering (status, payment, search)
`/api/reports/products`	Product analysis (top 10, categories, unsold)
`/api/reports/customers`	Customer analysis (top 50)
`/api/reports/compare`	Period comparison (sales growth, orders, distribution)
`/api/reports/offers`	Offer analysis (total discounts, top 10)
`/api/reports/invoices`	Invoice listing
`/api/reports/audit`	Status change audit log
`/api/reports/sales`	Sales analytics (Drizzle ORM)
`/api/reports/delivery-analytics`	Delivery analytics (fees by day/distance)
`/api/reports/cron`	Protected cron endpoint for scheduled reports
`/api/reports/schedule`	POST: Schedule email reports

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Section 6: Order Lifecycle

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```
┌──────────────────────────────────────────────────────────┐
│                   COMPLETE ORDER LIFECYCLE               │
└──────────────────────────────────────────────────────────┘

1. Browse Menu
   ├── Client: /menu → display items with prices
   ├── Client: useCartStore (Zustand, localStorage "bam-cart-storage")
   └── Support for bundles and individual offers

2. Create Order (/my-orders)
   ├── Client: Enter name, phone, address (required ≥10 chars)
   ├── Client: Select location on map (Leaflet)
   ├── Client: calculateDeliveryFeeServer(lat, lng) ← Server Action
   │   ├── Verify Sanaa location (isInsideSanaa)
   │   ├── Calculate OSRM distance (calculateRoute)
   │   └── Calculate delivery fee (calculateDeliveryFee)
   ├── Client: Submit order (createOrder)
   │
   └── Server Action: createOrder()
       ├── 1. Validate input data
       ├── 2. Fetch delivery settings (fetchDeliverySettings)
       ├── 3. Verify location (isInsideSanaa + max distance)
       ├── 4. Calculate distance (OSRM → Haversine fallback)
       ├── 5. Calculate delivery fee (calculateDeliveryFee)
       ├── 6. Generate order number (generateOrderNumber with Optimistic
Locking)
       ├── 7. Manage tracking token (get/create from customer_tokens)
       ├── 8. Validate offer (if applicable: offer validation + pricing)
       ├── 9. Save order (orders table + order_items)
       ├── 10. Save initial status (order_status_history)
       ├── 11. Save offer snapshot (order_offers + order_offer_items)
       └── 12. Return { order, trackingToken }

3. Track Order
   ├── Customer: /t/[token] → tracking page with QR + InvoiceModal
   └── Admin: /admin/orders → status management

4. Update Status (Admin)
   └── updateOrderStatus(orderId, newStatus, adminId?)
       ├── Read current status + version
       ├── Update with Optimistic Locking (.eq('version', currentVersion))
       ├── Log in order_status_history
       └── Handle concurrent modification conflicts

5. Invoice
   ├── Client: InvoiceModal → PNG (html2canvas) / PDF (jsPDF)
   └── Admin: receipt-html.ts → direct print (window.print)
```


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```
6. Archive
   └── is_archived = true / is_deleted = true (soft delete)
```


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Section 7: Delivery System

7.1 Delivery Fee Calculation Flow

```
Customer selects location on map
        |
        ▼
calculateDeliveryFeeServer(lat, lng)  ← Server Action
        |
        ├── isInsideSanaa(lat, lng)    ← Ray-casting on Sanaa boundaries
        |   └── × Outside Sanaa → Error
        |
        ├── fetchDeliverySettings()    ← site_settings (17 keys)
        |
        ├── haversineDistance()        ← Straight-line distance
        |
        ├── calculateRoute()           ← OSRM (8s timeout)
        |   ├── Success → validate with isRouteValid()
        |   ├── Failure → estimateRoadDistance(straightLine × roadFactor)
        |   └── Invalid → estimateRoadDistance(straightLine × roadFactor)
        |
        ├── Verify maxDistance
        |
        └── calculateDeliveryFee({distanceKm, settings})
            |
            ├── baseFee = baseDeliveryFee (if distance ≤ includedKm)
            |   OR baseFee = base + extraKm × extraFeePerKm
            |
            ├── + weatherFee if enabled
            |
            ├── + peakFee (baseFee × peakPercentage%) if:
            |   - Current time between peakStart and peakEnd
            |   - Current day is in peakDays
            |
            └── Return: {fee, baseFee, weatherFee, peakFee, ...}
```

7.2 Pricing Formula

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```
If distanceKm ≤ includedDistanceKm:
    deliveryFee = baseDeliveryFee
Otherwise:
    deliveryFee = baseDeliveryFee + (distanceKm - includedDistanceKm) ×
extraFeePerKm

Additional surcharges (if applicable):
    + weatherFee (if enabled)
    + baseFee × peakPercentage / 100 (if peak time and peak day)

Default values:
    baseDeliveryFee = 400 YER
    includedDistanceKm = 2 km
    extraFeePerKm = 100 YER/km
    maxDeliveryDistanceKm = 15 km
    weatherFee = 200 YER
    peakPercentage = 20%
```

7.3 OSRM Integration Flow

```
calculateRoute(originLat, originLng, destLat, destLng, straightLineKm,
roadFactor)
    |
    ├── URL: https://router.project-osrm.org/route/v1/driving/{lng},{lat};
{lng},{lat}
    |
    ├── Response: distance (meters), duration (seconds)
    |
    ├── isRouteValid(osrmKm, straightLineKm, roadFactor)
    |   ├── osrmKm < straightLine → false
    |   ├── osrmKm > straightLine × 10 → false
    |   └── osrmKm < straightLine × 0.75 → false
    |
    └── Fallback: straightLineKm × roadFactor (default 1.5)
```


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Section 8: Offer & Bundle System

8.1 Offer Types (4)

Type	Description	Example
`fixed_price`	Fixed price for bundle	"Family meal for 5000"
`percentage_discount`	Percentage discount	"20% off bundle"
`amount_discount`	Fixed amount discount	"1000 YER off"
`free_item`	Cheapest item free	"Buy 2 get 3rd free"

8.2 Pricing Engine (offer-pricing.ts)

```
calculateOfferPrice({ offerType, salePrice, discountPercent, discountAmount,
items })
    |
    ├── originalPrice = Σ(item.unitPrice × item.quantity)
    |
    ├── fixed_price         → finalPrice = salePrice ?? originalPrice
    ├── percentage_discount → finalPrice = originalPrice × (1 -
discountPercent/100)
    ├── amount_discount     → finalPrice = max(0, originalPrice -
discountAmount)
    └── free_item           → finalPrice = originalPrice -
unitPriceCheapestItem
    |
    └── return { originalPrice, discountAmount, discountPercent, finalPrice,
savings }
```

8.3 Snapshots

When creating an order with an offer:

1. `order_offers`: Stores `offer_name, original_price, discount_amount, final_price`
2. `order_offer_items`: Stores detailed bundle component data for each order

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3. This ensures report accuracy even if the offer changes later

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Section 9: Invoice System

9.1 Two Components, One Source

```
receipt-html.ts (Server/Print)
    ├── generateReceiptBody(order, settings, trackUrl, bundleInfo)
    │   └── HTML string → used in:
    │       ├── admin/reports/InvoicesTab.tsx (direct print)
    │       └── InvoiceModal.tsx (design matching)
    │
    └── generateReceiptHtml(order, settings, trackUrl, bundleInfo)
        └── Complete HTML document for printing
            └── Waits for images to load before window.print()

InvoiceModal.tsx (Client/Modal)
    ├── Display invoice in modal (React component)
    ├── Export PNG → html2canvas (scale: 3)
    └── Export PDF → jsPDF
        ├── format: [80mm, proportionalHeight]  ← single page
        └── addImage(imgData, 'PNG', 0, 0, 80mm, height)
```

9.2 Invoice Content Layout

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```
┌──────────────────────────┐
│    — Customer Invoice —   │
│  - - - - - - - - - - - - |
│        [Restaurant Logo]     |
│        Bait Al Mandi          |
│        Address                |
│    Phone: XXX – WhatsApp: XXX|
│  ──────────────────────── |
│  Invoice #    | BAM-XXXX..   |
│  Date         | June 15      |
│  Time         | 08:30 PM     |
│  Payment      | Cash         |
│  - - - - - - - - - - - - |
│  | Customer Info            |
│  | Name: Mohammed           |
│  | Phone: 777...            |
│  | Address: ...             |
│  ──────────────────────── |
│  | Order Details            |
│  | Item  | Size|Qty|Price|Tot|
│  | Mandi | Med | 2 |1500|3000|
│  - - - - - - - - - - - - |
│  | [Bundle Info if present]  |
│  ──────────────────────── |
│  Subtotal          3,000     |
│  Delivery Fee        400     |
│  - - - - - - - - - - - - |
│  Grand Total      3,400 YER  |
│  - - - - - - - - - - - - |
│       [QR Code for tracking] |
│  - - - - - - - - - - - - |
│     Thank you for your trust |
│         Bait Al Mandi        |
│  © 2026 Bait Al Mandi        |
└──────────────────────────┘
```

9.3 PDF Specifications

- **Width:** 80mm (standard thermal receipt)
- **Height:** Proportional to canvas (single page only)
- **Unit:** mm
- **Compression:** Yes (compress: true)
- **Margin:** 0 (no white borders)
- **Image Source:** Canvas from html2canvas at scale 3

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9.4 Print Specifications (Admin)

- **Size:** 320px width (centered on screen)
- **Fonts:** Tajawal (Google Fonts)
- **Colors:** `#3D0820` (maroon), `#c59b5f` (gold)
- **Image Loading:** JavaScript waits for all images before `window.print()`
- **@media print:** `page-break-inside: avoid` on all sections

9.5 Tracking Token System

```
Order Creation
    |
    ├── phone → search customer_tokens
    |   ├── exists → use existing token (stable for customer)
    |   └── not found → create new token (crypto.randomUUID)
    |
    ├── Store token in orders.tracking_token
    |
    ├── token ↔ QR Code on invoice
    |
    └── /t/[token] → order tracking page
        ├── Fetch order by tracking_token
        ├── Display status + details
        └── InvoiceModal with QR
```


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Section 10: Security & Authorization

10.1 Roles (3)

Role	Description	Permissions
`developer`	Full-access developer	Everything
`manager`	Restaurant manager	Dashboard, Menu, Categories, Offers, Gallery, Reviews, Reports, Settings, Delivery
`order_manager`	Order manager only	Orders only

10.2 Access Control

```
1. Middleware (src/middleware.ts)
   ├── Matches /admin/:path*
   ├── Calls updateSession from utils/supabase/middleware
   └── Verifies:
       ├── Auth session exists
       ├── admin_users record exists
       └── Page access permission (canAccessPage)

2. Server-Side (permissions.ts)
   ├── PAGE_ACCESS map
   ├── canAccessPage(role, pathname)
   ├── getAllowedSidebarLinks(role)
   └── getDefaultRedirect(role)

3. RLS (Row Level Security)
   └── rls.sql contains:
       ├── is_admin()        ← Check if admin in admin_users
       ├── get_admin_role()  ← Get admin role
       ├── Public SELECT on: categories, items, item_prices, offers,
       │                     gallery_images, site_settings, branches
       ├── Admin-only INSERT/UPDATE/DELETE
       ├── Customers see only their orders (customer_phone)
       └── Admins see all orders
```


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10.3 Server Action Protection

- `createOrder` → `'use server'` → validates data only (not Auth)
- `updateOrderStatus` → `'use server'` → no direct Auth check (used by Admin)
- `calculateDeliveryFeeServer` → `'use server'` → no Auth (public service)
- Permission is enforced at middleware + sidebar level

10.4 Potential Vulnerabilities

Vulnerability	Status	Impact
`updateOrderStatus` without Auth check	⚠️ Depends on Middleware	Low - callable from any Client
`createOrder` without CSRF protection	⚠️ No CSRF Token	Medium
`calculateDeliveryFeeServer` public	✅ Intentional	Low - preview only
Tracking token guessable	✅ `crypto.randomUUID()`	Low
`site_settings` public read	✅ Intentional	Low

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Section 11: Reporting System

11.1 Architecture

```
Report Interface (Client)
    └── /admin/reports/page.tsx
        └── 9 Tabs
            ├── DashboardTab
            ├── OrdersTab
            ├── ProductsTab
            ├── CustomersTab
            ├── SummaryTab
            ├── OffersTab
            ├── InvoicesTab
            ├── AuditLogsTab
            └── DeliveryAnalyticsTab
                │
                └── API Routes (12 endpoints)
                    ├── REST (Supabase Client)
                    └── Drizzle ORM (server-side)
                        └── services/reports/
                            ├── salesReport.ts
                            ├── analyticsEngine.ts
                            ├── customerReport.ts
                            ├── dashboardReport.ts
                            ├── ordersReport.ts
                            └── productReport.ts
```

11.2 Print & Export

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```
Print (printReport.ts)
    ├── fetchData() → collect data from API
    ├── buildSection() → build complete HTML with inline CSS
    ├── Open print window (window.open)
    └── wait + window.print()

Excel Export (exportExcel.ts)
    └── ExcelJS → multi-sheet .xlsx
        ├── Orders Sheet
        ├── Products Sheet
        ├── Customers Sheet
        ├── Sales Comparison Sheet
        └── Delivery Analytics Sheet
```


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Section 12: State Management (Stores)

12.1 cartStore (Zustand + persist)

```
// localStorage key: "bam-cart-storage"
// Middleware: persist

useCartStore(
    state: {
        items: CartItem[]   // [id, name, price, quantity, size, category,
image, offer...]
    },
    actions: {
        addToCart(item)     // Search for duplicate (id + size) → increment
quantity
        removeFromCart(id)  // Remove item
        updateQuantity(id, quantity)  // Change quantity (0 → remove)
        clearCart()         // Empty cart
        getCartTotal()      // Σ(price × quantity)
        getTotalItems()     // Σ(quantity)
    }
)
```

12.2 SettingsContext (React Context)

```
SettingsProvider
    ├── refresh() → fetch site_settings from Supabase
    ├── Transform data to Record<string, string>
    └── Provide { settings, loading, refresh } to children

useSettings() → { settings: Record<string, string>, loading: boolean }
```


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Section 13: Legacy Files (Unused)

These files were part of the old zone-based pricing system and are no longer imported by any production code:

File	Reason
`src/lib/delivery-zones.ts`	Zone system (Azal, Shattar, etc.) — replaced by distance pricing
`src/lib/geo/resolve-zone.ts`	Zone resolution via Nominatim API — no longer used
`src/lib/geo/resolve-location.ts`	R-Tree spatial indexing engine — no longer used
`src/lib/geo/sanaa-boundaries.ts`	GeoJSON loading — no longer used
`src/app/api/resolve-zone/route.ts`	Zone resolution API — no longer used
`src/app/cart/page.tsx`	Old cart page — replaced by `/my-orders`
`src/app/track-order/[orderId]/page.tsx`	UUID-based tracking — replaced by `/t/[token]`
`delivery_zone` column in orders	Always NULL

Note: The `delivery_zones` table still exists in the database but is unused in any code. `delivery_zone: null` is set in `createOrder` for backward compatibility.

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Section 14: Dependency Map

```
components/ ← app/ ← layout.tsx ← lib/settings-context.tsx ← lib/supabase.ts
                                                  |
actions/orders.ts ──────────────────────────┬── lib/delivery-pricing.ts
    |                                       |── lib/delivery-routing.ts
    |                                       |── lib/location-validation.ts
    |                                       └── lib/offer-pricing.ts
    |
actions/items.ts ──────────────────────────── lib/offer-pricing.ts
actions/categories.ts
actions/orders-offers.ts ──────────────────── db/schema.ts (types)

store/cartStore.ts ────────────────────────── (independent, Zustand)

app/admin/ ──────────────────────────────── middleware.ts
    |                                          utils/supabase/middleware.ts
    |                                          lib/permissions.ts
    |
app/admin/reports/ ──────────────────────── services/reports/
    |                                          db/index.ts (Drizzle)
    |                                          lib/printReport.ts
    |                                          lib/exportExcel.ts
    |
components/invoice/InvoiceModal.tsx ──────── lib/bundle-utils.ts
    |                                          actions/orders-offers.ts
    |
components/invoice/receipt-html.ts ──────── lib/bundle-utils.ts

API Routes (/api/reports/*) ──────────────── services/reports/
                                                 db/index.ts (Drizzle)
                                                 lib/supabase.ts
```


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Section 15: Executive Summary

15.1 Strengths

- ✅ **Single Source of Truth** for delivery fees (`calculateDeliveryFee`)
- ✅ **All Calculations Server-Side** (client values never trusted)
- ✅ **Optimistic Locking** for orders (prevents concurrent updates)
- ✅ **Snapshots for offers and delivery** (ensures report accuracy)
- ✅ **4 Offer Types** (fixed, percentage, amount, free)
- ✅ **3-Way Tracking** (Admin UI, Token Link, QR Code)
- ✅ **3 Format Invoices** (PNG, PDF, Direct Print)
- ✅ **Complete Reporting System** (9 tabs + Print + Excel)
- ✅ **Extensible Database** (Drizzle ORM + SQL Migrations)
- ✅ **3 Roles** (Developer, Manager, Order Manager)

15.2 Recommended Improvements

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Issue	Severity	Recommendation
`updateOrderStatus` without CSRF/Token	Medium	Add session verification inside Server Action
`createOrder` without Auth	Medium	Add Rate Limiting + additional validation
`fetchDeliverySettings` called twice (preview + creation)	Low	Merge the two calls or use caching
Legacy files not deleted	Low	Delete unused files (delivery-zones.ts, geo/*, cart/page.tsx)
Tracking Token persists forever	Low	Add token expiry
`services/reports/` uses Drizzle but some API routes use Supabase REST	Low	Unify approach (all Drizzle or all REST)
No DB Migration to drop `delivery_zones`	Low	Add migration to drop the table

15.3 Project Deliverables

File	Purpose
`BAIT_AL_MANDI_ARCHITECTURE.md`	Complete system documentation (Markdown)
`exports/BAIT_AL_MANDI_ARCHITECTURE_AR.pdf`	Arabic PDF documentation
`exports/BAIT_AL_MANDI_ARCHITECTURE_EN.pdf`	English PDF documentation

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